



PROGRAMING FUNDAMENTALS

PROJECT REPORT

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Project Title:

“JAVA QUIZ APPLICATION”

1. Introduction :

The **Quiz Application** is a Java-based console program designed to facilitate quiz-taking for users and quiz management for administrators. The system allows users to register, log in, and attempt quizzes based on different difficulty levels (Easy, Medium, Hard, and Surprise). Administrators can add, edit, or delete quizzes and view all user results.

The application features:

- **User Registration & Login** – Ensures secure access.
- **Quiz Difficulty Levels** – Users can choose from Easy, Medium, Hard, or a Surprise quiz.
- **Timed Quizzes** – Each difficulty level has a predefined time limit.
- **Admin Panel** – Allows quiz management (add, edit, delete) and result tracking.
- **Result Tracking** – Users can view their past results, while admins can see all results.

The system is file-based, storing user registrations, quiz questions, answer keys, and results in text files.

2. Task Distribution:

TASKS	DONE BY
• User Registration & Login	Zabih Ullah
• Quiz Taking Module (Difficulty Levels, Timer)	Masab Haroon
• Admin Module (Add/Edit/Delete Quizzes)	Masab Haroon
• Result Tracking (User & Admin Views)	Zabih Ullah
• File Handling (User Data, Quiz Questions, Results)	Masab Haroon
• Code and output Formatig	Zabih Ullah
• Testing & Debugging	Both

3. Features of the System:

- **User Features:**

- Registration with age validation (must be 15+).
- Secure login with password strength checks.
- Multiple quiz difficulty levels (Easy, Medium, Hard, Surprise).
- Timed quizzes with countdown display.
- View personal quiz results.
- **Admin Features:**
 - Add new quizzes (subject + difficulty).
 - Edit existing quizzes (add questions).
 - Delete quizzes.
 - View all users' results.
- **Technical Features:**
 - File-based data storage (registrations.txt, adminlogin.txt, AllResults.txt).
 - Input validation for user choices.
 - Password strength enforcement (uppercase, digits, special characters).

4. Modules or Entities:

1. User Authentication Module

- **Registration:** Users provide a username, age (must be 15+), and a strong password.
- **Login:** Users log in with credentials; three attempts allowed.

2. Quiz Module

- **Difficulty Selection:** Users choose between Easy, Medium, Hard, or Surprise.
- **Timed Quiz:** Each difficulty has a different time limit (Easy: 2.5 min, Hard: 10 min, Random time).
- **Question Format:** MCQs stored in files (e.g., MathEasyQuiz.txt with corresponding keys in MathEasyKey.txt).

3. Admin Module

- **Add Quiz:** Admins create new quiz files for a subject and difficulty.
- **Edit Quiz:** Admins append new questions to existing quizzes.
- **Delete Quiz:** Removes quiz and key files, updates subject list.

4. Results Module

- **User Results:** Stored in [Username].txt.
- **Admin Results:** All results stored in AllResults.txt.

Appendix: Files and Methods in Quiz Application

1. Files Used in the System:

1. registrations.txt

- Purpose: Stores user credentials
- Format: username|password|age
- Example: johnDoe|Pass123!|20

2. adminlogin.txt

- Purpose: Stores admin login credentials
- Format: username|password
- Example: admin|Admin@123

3. Subjects.txt

- Purpose: Lists available quiz subjects with numbering
- Format: | 1: Math |
- Example: | 1: Math |\n| 2: Science |

4. [Subject][Difficulty]Quiz.txt

- Purpose: Contains MCQ questions
- Format: Question \$A. \$B. Option \$C. Option \$D. Option
- Example: MathEasyQuiz.txt with content: What is 2+2? \$A. 3\$B. 4\$C. 5\$D. 6

5. [Subject][Difficulty]Key.txt

- Purpose: Contains correct answers
- Format: Single character (A/B/C/D) per line
- Example: MathEasyKey.txt with content: B

6. AllResults.txt

- Purpose: Stores all quiz attempts
- Format: [Username] scored [X]/[Y] in [Subject][Difficulty]
- Example: johnDoe scored 4/5 in MathEasy

7. [Username].txt

- Purpose: Stores individual user's quiz results
- Example: johnDoe.txt containing past quiz results
- 8. **SurpriseQuiz.txt**
 - Purpose: Contains random difficulty questions
 - Format: Same as regular quiz files
- 9. **SurpriseKey.txt**
 - Purpose: Contains answers for surprise quiz
 - Format: Same as regular key files

2. Key Methods in the Code

1. **isStrong(String password)**
 - Validates password meets requirements
 - Checks for: 8+ characters, 1 uppercase, 1 digit, 1 special character
 - Returns: boolean (true if strong)
2. **duplicateUsers(String userName)**
 - Checks if username exists in registrations.txt
 - Returns: boolean (true if duplicate exists)
3. **duplicateUsers(String userName, String password)**
 - Checks if user credentials already exist
 - Used for login validation
4. **registerUser()**
 - Handles new user registration
 - Writes to registrations.txt
 - Includes age validation (15+ years)
5. **loginUser()**
 - Authenticates users against registrations.txt
 - Allows 3 attempts before locking out
6. **loginAdmin()**
 - Authenticates admins against adminlogin.txt
 - Also allows 3 attempts
7. **difficultyLevel()**
 - Presents difficulty options menu
 - Options: Easy, Medium, Hard, Surprise

- Returns selected difficulty as String

8. startQuiz()

- Manages quiz selection flow
- Shows subject menu
- Calls takeQuiz() with appropriate files

9. takeQuiz(String questionFile, String key, String userName, int time)

- Core quiz engine
- Features:
 - Loads questions from file
 - Manages countdown timer
 - Checks answers against key file
 - Records results to user file and AllResults.txt

10.viewResults()

- Displays user's past results
- Reads from [Username].txt
- Shows formatted results

11.viewAllResults()

- Displays all users' results
- Reads from AllResults.txt
- Admin-only function

12.addQuiz()

- Admin function to create new quizzes
- Creates:
 - New quiz file ([Subject][Difficulty]Quiz.txt)
 - Corresponding key file
- Updates Subjects.txt

13.editQuiz()

- Admin function to modify quizzes
- Appends new questions to existing files
- Maintains format consistency

14.deleteQuiz()

- Admin function to remove quizzes

- Deletes:
 - Quiz file
 - Key file
- Updates Subjects.txt

15.lineCounter(String File)

- Helper method
- Counts lines in a file
- Used for tracking total questions in a quiz

➤ SOURCE CODE:

```

1  import java.util.*;
2  import java.io.*;
3  public class QuizApplication {
4      static Scanner input = new Scanner(System.in);
5      static boolean isUserLogin = false;
6      static boolean isAdminLogin = false;
7      static int age;
8      static String currentUser ;
9      public static boolean isStrong(String password){
10         if (password.length() < 8){
11             return false;
12         }
13         boolean hasDigit = false;
14         boolean hasUpper = false;
15         boolean hasLetter = false;
16         boolean hasSpecial = false;
17
18         //loop for checking the password
19         for(int i = 0 ; i < password.length() ; i++){
20             //storing character of password to check
21             char character = password.charAt(i);
22             if(Character.isDigit(character)){
23                 hasDigit = true;
24             }
25             if(Character.isLetter(character)){
26                 hasLetter = true;
27             }
28             if(Character.isUpperCase(character)){
29                 hasUpper = true;
30             }
31             if (!(Character.isDigit(character) && Character.isLetter(character))) {
32                 hasSpecial = true;
33             }
34         }
35         return hasDigit && hasLetter && hasSpecial && hasUpper;
36     }
37     public static boolean duplicateUsers(String userName , String password){
38
39         //opening the registration file
40         File file = new File(pathname:"registrations.txt");
41         try{
42             Scanner reader = new Scanner(file);
43             while (reader.hasNext()){
44                 String line = reader.nextLine();
45                 String[] registrationDetails = line.trim().split(regex:"\\|");

```

```

46
47         if(userName.equals(registrationDetails[0]) &&
48             | password.equals(registrationDetails[1])){
49             |     return true;
50         }
51     }
52     reader.close();
53 }
54 catch (FileNotFoundException e){
55     System.out.println(x:"File Does Not Exists!");
56 }
57 return false;
58 }
59 public static boolean duplicateUsers(String userName ){
60
61     //opening the registration file
62     File file = new File(pathname:"registrations.txt");
63     try{
64         Scanner reader = new Scanner(file);
65         while (reader.hasNext()){
66             String line = reader.nextLine();
67             String[] registrationDetails = line.trim().split(regex:"\\|");
68
69             if(userName.equals(registrationDetails[0])){
70                 |     return true;
71             }
72         }
73         reader.close();
74     }
75     catch (FileNotFoundException e){
76         System.out.println(x:"File Does Not Exists!");
77     }
78     return false;
79 }
80 public static void registerUser() throws InputMismatchException{
81     String userName;
82     String password;
83
84
85     while (true) {
86         //prompting user to enter user-name & password
87         System.out.print(s:" >> Enter Your Name: ");
88         userName = input.nextLine();
89
90         //checking if user-name already taken
91         if(duplicateUsers(userName)){
92             System.out.println(x:"> User Name Already Taken");
93             System.out.println(x:" Kindly Use Another User-Name");
94             continue;
95         }
96         while (true) {
97             try {
98                 System.out.print(s:" >> Enter Your Age: ");
99                 age = input.nextInt();
100                 break;

```



```

101 } catch (InputMismatchException e) {
102     System.out.println(x:"> Only Integers are Accepted....Try Again");
103     input.nextLine();
104 }
105 }
106
107 //validating age
108 if( age < 15){
109     return;
110 }
111 //clearing buffer
112 input.nextLine();
113 System.out.print(s:" >> Enter a strong Password: ");
114 password = input.nextLine();
115 //checking if user already exists
116 if(duplicateUsers(userName,password)){
117     System.out.println(x:">> User Already Exists Kindly Login");
118     return;
119 }
120
121 if (!isStrong(password)) {
122     System.out.println(x:" Weak Password! .... Kindly Enter a Strong Password");
123 }
124 else {
125     //if password is strong then exit the loop
126     System.out.println(x:"***** <Registered Successfully> *****");
127     break;
128 }
129 }
130 try {
131     // opening file for storing registration records
132     FileWriter writer = new FileWriter(fileName:"registrations.txt", append:true);
133     writer.write(userName + "|" + password + "|" + age + "\n");
134     //closing writer
135     writer.close();
136 } catch (FileNotFoundException e){
137     System.out.println(x:"File Does Not Exist");
138 }
139 catch (IOException e){
140     System.out.println(x:"Error While Writing to the file");
141 }
142 }
143 public static void loginUser() {
144     int attempts = 3;
145     while (true) {
146         System.out.print(s:">> UserName: ");
147         String userName = input.nextLine();
148
149         while (true) {
150             try {
151                 System.out.print(s:">> Age: ");
152                 age = input.nextInt();
153                 break;
154             } catch (InputMismatchException e) {
155                 System.out.println(x:"Only Integers are Accepted....Try Again");
156                 input.nextLine();
157             }
158         }
159
160         //clearing buffer

```

```

186         System.out.println(x: " -----");
187         System.out.println(x: " |    Better Luck Next Time!!    |");
188         System.out.println(x: " | You Have Wasted All Your Attempts |");
189         System.out.println(x: " -----");
190
191         break;
192     }
193     if (isUserLogin) {
194         System.out.println(x: "***** User Login Successful *****");
195         break;
196     } else {
197         System.out.println(x: "Invalid Credentials. Try Again!");
198         System.out.println("> Attempts left: " + attempts);
199     }
200 }
201 }
202 }
203 public static void loginAdmin() {
204     int attempts = 3;
205
206     do{
207         System.out.print(s: ">> UserName: ");
208         String userName = input.nextLine();
209         System.out.print(s: ">> Password: ");
210         String password = input.nextLine();
211
212         //opening the registration file to read data so to verify login
213         File file = new File(pathname:"adminlogin.txt");
214         try {
215             Scanner reader = new Scanner(file);
216             while (reader.hasNext()) {
217                 //storing line from the file in a variable
218                 String line = reader.nextLine();
219                 String[] details = line.trim().split(regex:"\\|");
220
221                 //checking if the login details matches
222                 if (userName.equals(details[0]) && password.equals(details[1])) {
223                     isAdminLogin = true;
224                 }
225             }
226             reader.close();
227         } catch (FileNotFoundException e) {
228             System.out.println(x: "Can not Find the File");
229         }
230     } if (isAdminLogin) {
231         System.out.println(x: "***** Admin Login Successfully *****");
232
233         //getting out of loop if login successfully
234         break;
235     }
236
237     else {
238         System.out.println(x: "Invalid Credentials ..... Try Again!");
239         System.out.println("You Have " + (attempts - 1) + ((attempts > 2)? " attempts":" attempt") + " left");
240     }
241     attempts --;
242 }while (attempts != 0 );
243 }
244 public static void showUserQuizMenu() {
245     System.out.println(x: "=====");

```

```

Click to add a breakpoint
246 System.out.println(x:"|    << Welcome to User Quiz Section >>    |");
247 System.out.println(x:"=====");
248 System.out.println(x:"| 1: Take Quiz |");
249 System.out.println(x:"| ----- |");
250 System.out.println(x:"| 2: View Results |");
251 System.out.println(x:"| ----- |");
252 System.out.println(x:"| 3: Exit |");
253 System.out.println(x:"=====");
254
255 //variable that will store user selected option
256 int userMenuOption;
257
258 while (true) {
259     try {
260         System.out.print(s:">> Kindly Select an Option: ");
261         userMenuOption = input.nextInt();
262
263         if (userMenuOption < 1 || userMenuOption > 3){
264             System.out.println(x:"Invalid Option....Try Again");
265             continue;
266         }
267         //getting out of loop when valid input received
268         break;
269     } catch (InputMismatchException e) {
270         System.out.println(x:"> Only Integers are Allowed! .... Try Again");
271         input.nextLine();
272     }
273 }
274 //checking what user selected from the MENU
275 switch (userMenuOption) {
276
277     case 1 -> startQuiz();
278
279     case 2 -> viewResults();
280
281     case 3 -> {
282         System.out.println(x:"> Logging Out.....");
283     }
284 }
285 }
286 public static void showAdminQuizMenu() {
287
288     System.out.println(x:"=====");

```

```

289 System.out.println(x:"| Welcome to Admin Panel |");
290 System.out.println(x:"=====");
291 System.out.println(x:"| 1: Add Quiz |");
292 System.out.println(x:"| ----- |");
293 System.out.println(x:"| 2: View All Results |");
294 System.out.println(x:"| ----- |");
295 System.out.println(x:"| 3: Edit Quiz |");
296 System.out.println(x:"| ----- |");
297 System.out.println(x:"| 4: Delete Quiz |");
298 System.out.println(x:"| ----- |");
299 System.out.println(x:"| 5: Exit |");
300 System.out.println(x:"=====");
301

```

```

301
302     int adminChoice = 0;
303
304     while (true) {
305         try {
306             System.out.print(s:">> Kindly Select an option: ");
307             adminChoice = input.nextInt();
308             if(adminChoice < 1 || adminChoice > 5){
309                 System.out.println(x:"> Invalid Choice.... Try Again");
310                 continue;
311             }
312             //clearing buffer
313             input.nextLine();
314             //getting out of loop if correct input is provided
315             break;
316         } catch (InputMismatchException e) {
317             System.out.println(x:"> Only Integers are Accepted....Try Again");
318             input.nextLine();
319         }
320     }
321     switch (adminChoice) {
322         case 1 -> addQuiz();
323
324         case 2 -> viewAllResults();
325
326         case 3 -> editQuiz();
327
328         case 4 -> deleteQuiz();
329
330         case 5 -> {
331             System.out.println(x:"Logging Out....");
332             isAdminLogin = false;
333         }
334     }
335 }
336 public static void subjects(){
337     File file = new File(pathname:"Subjects.txt");
338     try{
339         Scanner reader = new Scanner(file);
340         while (reader.hasNext()){
341             String line = reader.nextLine();
342             System.out.println(line);
343         }
344     }catch (FileNotFoundException e){
345         System.out.println(x:"File Does Not Exists");
346     }
347 }
348 public static void startQuiz() {
349     char userWants;
350     //checking what difficulty level is selected by the user
351     String difficulty = difficultyLevel().toUpperCase();
352
353     //setting timer for the quiz based on the difficulty

```

```

353     //setting timer for the quiz based on the difficulty
354     int totalTimeInSeconds = switch (difficulty.toUpperCase()){
355         //setting quiz time 2.5 minutes if user selected easy
356         case "EASY" -> 150;
357         //setting quiz time 5 minutes if user selected medium
358         case "MEDIUM" -> 5 * 60;
359         //setting quiz time 10 minutes if user selected hard
360         case "HARD" -> 10 * 60;
361         //setting timer randomly between 1 - 10
362         case "SURPRISE" -> (int)(1 + Math.random() * 10 ) * 60;
363         default -> 300;
364     };
365
366     if(difficulty.equals(anObject:"SURPRISE")){
367         takeQuiz(questionFile:"SurpriseQuiz.txt",key:"SurpriseKey.txt",currentUser,totalTimeInSeconds);
368         System.out.print(s:">> Want to Go Back to Menu (Y/N)?");
369         userWants = input.next().toUpperCase().charAt(index:0);
370         if(userWants == 'Y')

```

```

370         if(userWants == 'Y'){
371             showUserQuizMenu();
372             return;
373         }
374         else {
375             System.out.println(x:"Logging Out.....");
376             return;
377         }
378     }
379
380     // if user selects surprise then quiz menu will not be shown to the user
381     System.out.println(x:"===== QUIZ MENU =====");
382     subjects();
383     System.out.println(x:"=====");
384
385     //variable that will store what user selected from the QUIZ-MENU
386     String subjectName;
387     input.nextLine();
388     System.out.print(s:"Enter the Subject Name to Attempt Quiz: ");
389     subjectName = input.nextLine();
390
391     String questionFile = subjectName + difficulty + "Quiz.txt";
392     String keyFile = subjectName + difficulty + "Key.txt";
393
394     takeQuiz(questionFile,keyFile,currentUser,totalTimeInSeconds);
395
396     System.out.println("Do You Want to Go to back to MENU then press Y otherwise " +
397         | | | "press any other key to exit");
398     userWants = input.next().toUpperCase().charAt(index:0);
399     if (userWants != 'Y') {
400         System.out.println(x:"Logging Out.....");
401     }
402     else {
403         showUserQuizMenu();
404     }
405 }
406
407 //method to get total question that will help in compiling the result
408 public static int lineCounter(String File){
409     int counter = 0;
410     File file = new File(File);
411     try{
412         Scanner reader = new Scanner(file);
413         while (reader.hasNext()){
414             String line = reader.nextLine();
415             counter ++;
416         }
417         reader.close();
418     }catch (FileNotFoundException e){
419         System.out.println(x:"File Does Not Exists");
420     }
421     return counter;
422 }

```

```

422 public static void takeQuiz(String questionFile , String key,String userName, int totalTimeInSeconds) {
423     int score = 0;
424     //getting how many total questions are there in the file
425     int questions = lineCounter(questionFile);
426
427     File file = new File(questionFile);
428     File file2 = new File(key);
429
430     try {
431         //for writing results in user file
432         FileWriter writer = new FileWriter(userName + ".txt", append:true);
433
434         //for writing result in a file accessible by admin
435         FileWriter resultWriter = new FileWriter(fileName:"AllResults.txt",append:true);
436         // for reading questions
437         Scanner reader = new Scanner(file);
438         // for reading key
439         Scanner keyReader = new Scanner(file2);
440

```

```

441 // Start timer
442 long startTime = System.currentTimeMillis();
443 long lastMinuteUpdate = 0;
444
445 //giving user message that how much time user have to attempt the quiz
446 System.out.println(x:"===== NOTE! =====");
447 System.out.println("| You Have " + (totalTimeInSeconds/60) + " minutes and "
448 | + (totalTimeInSeconds % 60) + " seconds to complete the Quiz |");
449 System.out.println(x:"=====");
450 while (reader.hasNext() && keyReader.hasNext()) {
451
452     //checking how much time has passed
453     long elapsedTime = (System.currentTimeMillis() - startTime) / 1000;
454     //checking how much time is left
455     long timeLeft = totalTimeInSeconds - elapsedTime;
456
457     if (timeLeft <= 0) {
458         System.out.println(x:"\n Time's up! Quiz ended automatically.");
459         break;
460     }
461
462     // Showing user how much time is left for completing the quiz
463     if (elapsedTime - lastMinuteUpdate >= 30) {
464         System.out.println(" Time Left: " + (timeLeft / 60) + " minutes " + (timeLeft % 60) + " seconds");
465         lastMinuteUpdate = elapsedTime;
466     }
467
468     String line = reader.nextLine();
469     String[] mcq = line.trim().split(regex:"\\$");
470
471     char correctAnswer = keyReader.next().charAt(index:0);
472
473     for (String question : mcq) {
474         System.out.println(question);
475     }
476     char userAnswer;
477     while (true) {
478         System.out.print(s:"Enter Your Answer: ");
479         userAnswer = input.next().toUpperCase().charAt(index:0);
480         if (userAnswer != 'A' && userAnswer != 'B'
481             && userAnswer != 'C' && userAnswer != 'D'){
482             System.out.println(x:"Select option(A,B,C or D)");
483         }
484         else {
485             //clearing buffer
486             input.nextLine();
487             break;
488         }
489     }
490 }
491
492 if (userAnswer == correctAnswer) {
493     score++;
494 }
495 }
496
497 System.out.println("You Scored " + score + " out of " + questions);
498
499 //writing data in user results file
500 writer.write(userName + " Scored " + score + " out of " + questions + " in "
501 | + questionFile.substring(beginIndex:0, questionFile.indexOf(str:".")) + "\n");
502
503 //writing data in allResults file
504 resultWriter.write(userName + " scored " + score + "/"
505 | + lineCounter(questionFile) + " in " + questionFile.substring(beginIndex:0,questionFile.indexOf(str:"Q")) + "\n" );
506
507 writer.close();
508 reader.close();
509 keyReader.close();
510 resultWriter.close();
511 } catch (FileNotFoundException e) {
512     System.out.println(x:"File Not Found Exception");
513 } catch (IOException e) {
514     System.out.println(x:"Error while reading/writing the file");
515 }

```

```

517 public static void viewResults() {
518     char userWants;
519     //opening the file in which result is stores
520     File file = new File(currentUser + ".txt");
521     if(!file.exists()){
522         System.out.println("> There is no Previous Record of " + currentUser);
523         System.out.println(x:"THANKS!");
524
525         System.out.print(s:">> Do You Want to go back to USER-MENU(Y/N)? : ");
526         userWants = input.next().toUpperCase().charAt(index:0);
527         if(userWants != 'Y'){
528             System.out.println(x:"Logging Out.....");
529             isUserLogin = false;
530         }
531         else {
532             showUserQuizMenu();
533         }
534         return;
535     }
536     try{
537         //creating scanner object to read from the file
538         Scanner reader = new Scanner(file);
539         while (reader.hasNext()){
540             String line = reader.nextLine();
541             System.out.println(line);
542         }
543         reader.close();
544     }
545     catch (FileNotFoundException e){
546         System.out.println(x:"File Does Not Exist!");
547     }
548     System.out.print(">> Do You Want to Go to back to MENU then press Y otherwise " +
549         "press any other key to exit: ");
550     userWants = input.next().toUpperCase().charAt(index:0);
551     if (userWants != 'Y') {
552         System.out.println(x:"Logging Out.....");
553         isUserLogin = false;
554     }
555     else {
556         showUserQuizMenu();
557     }
558 }
559 public static void viewAllResults(){
560     File file = new File(pathname:"AllResults.txt");
561     try{
562         Scanner reader = new Scanner(file);
563         while (reader.hasNext()){
564             String line = reader.nextLine();
565             System.out.println(line);
566         }
567     }catch (IOException e){
568         System.out.println(x:"Error While Reading data from the file");
569     }
570 }
571
572 System.out.print(">> Do You Want to Go to back to MENU then press Y otherwise " +
573     "press any other key to exit: ");
574 char adminWants = input.next().toUpperCase().charAt(index:0);
575 if (adminWants != 'Y') {
576     System.out.println(x:"Logging Out.....");
577     isAdminLogin = false;
578 }
579 }
580 public static void addQuiz() {
581     System.out.print(s:"Enter the Subject Name to Add: ");

```

```

580 public static void addQuiz() {
581     System.out.print(s:"Enter the Subject Name to Add: ");
582     String subjectName = input.nextLine();
583
584     //Now allowing admin to select the difficulty
585     System.out.println(x:"=====");
586     System.out.println(x:"| 1: EASY |");
587     System.out.println(x:"| 2: MEDIUM |");
588     System.out.println(x:"| 3: HARD |");
589     System.out.println(x:"=====");
590     int difficultyOption;
591     while (true){
592         try{
593             System.out.printf(format:">> Select a Difficulty Level for %s",subjectName);
594             difficultyOption = input.nextInt();
595             if(difficultyOption < 1 || difficultyOption > 3){
596                 System.out.println(x:"> Kindly Select a Valid Option");
597             }
598             else {
599                 break;
600             }
601         }catch (InputMismatchException e){
602             System.out.println(x:"Only Integers Are Accepted.....Try Again");
603         }
604     }
605     String difficulty = "";
606     switch (difficultyOption){
607         case 1 -> difficulty = "EASY";
608         case 2 -> difficulty = "MEDIUM";
609         case 3 -> difficulty = "HARD";
610     }

```

```

611 int lineNumber = lineCounter(file1"Subjects.txt");
612
613
614 //creating files
615 File file1 = new File(subjectName + difficulty + "Quiz.txt");
616 File file2 = new File(subjectName + difficulty + "Key.txt");
617
618 try {
619     //creating file writer Object to append data
620     FileWriter writer = new FileWriter(fileName:"Subjects.txt",append:true);
621
622     //creating Print Writer so we can format data in a specific way
623     PrintWriter writer1 = new PrintWriter(writer);
624
625     //counting line number in subject file
626     lineNumber++;
627
628     writer1.printf(format:"%d: %s\n",lineNumber,subjectName);
629
630     System.out.println(subjectName + "- Added to the Subjects Successfully!");
631     if(file1.createNewFile()){
632         System.out.println("-> Quiz File " + file1.getName() + " Created Successfully");
633     }
634     else {
635         System.out.println("-> Quiz File " + file1.getName() + " Already Exists");
636     }
637     if(file2.createNewFile()){
638         System.out.println("-> Key File " + file2.getName() + " Created Successfully");
639     }
640     else {
641         System.out.println("-> Key File " + file2.getName() + " Already Exists");
642     }
643     writer.close();
644     writer1.close();
645 } catch (FileNotFoundException e){
646     System.out.println("File Does Not exists");
647 }
648 catch (IOException e){
649     System.out.println("Error Occurred Writing or creating the files");
650 }
651
652 char adminWants;
653 System.out.print("> Do You Want to Go to back to MENU then press Y otherwise " +
654     "press any other key to exit: ");
655 adminWants = input.next().toUpperCase().charAt(index:0);
656 if (adminWants != 'Y') {
657     System.out.println("Logging Out.....");
658     isAdminLogin = false;
659 }
660
661 public static void deleteQuiz(){
662     System.out.print("Enter the Subject Name to Delete: ");
663     String subjectName = input.nextLine();
664
665     //Now allowing admin to select the difficulty
666     System.out.println("-----");
667     System.out.println("1: EASY");
668     System.out.println("2: MEDIUM");
669     System.out.println("3: HARD");
670     System.out.println("-----");
671     int difficultyOption;
672     while (true){
673         try{
674             System.out.printf(format:"Select a Difficulty Level for: %s\n",subjectName);
675             difficultyOption = input.nextInt();
676             if(difficultyOption < 1 || difficultyOption > 3){
677                 System.out.print("Kindly Select a Valid Option: ");
678             }
679             else {
680                 break;
681             }
682         } catch (InputMismatchException e){
683             System.out.println("Only Integers Are Accepted.....Try Again");
684         }
685     }
686     String difficulty = "";
687     switch (difficultyOption){
688         case 1 -> difficulty = "EASY";
689         case 2 -> difficulty = "MEDIUM";

```

```

691     }
692     String quizFile = subjectName + difficulty + "Quiz.txt";
693     String keyFile = subjectName + difficulty + "Key.txt";
694
695     File file1 = new File(quizFile);
696     File file2 = new File(keyFile);
697
698     if(file1.delete() && file2.delete()){
699         System.out.println("All the files of " + subjectName + " deleted Successfully!");
700     }
701
702     //now logic for updating the subjects file
703     File subjects = new File(pathname:"Subjects.txt");
704     File tempFile = new File(pathname:"temp.txt");
705
706     try{
707         tempFile.createNewFile();
708         Scanner reader = new Scanner(subjects);
709         PrintWriter writer = new PrintWriter(tempFile);
710

```



```

711         while (reader.hasNext()) {
712             String line = reader.nextline();
713             if (line.contains(subjectName)) {
714                 writer.println(line);
715             }
716         }
717         reader.close();
718         writer.close();
719     } catch (IOException e) {
720         System.out.println("Error while reading or writing to the file");
721     }
722
723     //now deleting the subject file & renaming temp file with subject file
724     if(subjects.delete() && tempFile.renameTo(subjects)){
725         System.out.println(subjectName + " -> Subject removed from the Subject File Successfully");
726     }
727     else {
728         System.out.println("Error while updating the subjects file");
729     }
730
731     char adminWants;
732     System.out.print("Do You Want to Go to back to MENU then press Y otherwise " +
733         " | " + "press any other key to exit: ");
734     adminWants = input.next().toUpperCase().charAt(index:0);
735     if (adminWants != 'Y') {
736         System.out.println("logging Out.....");
737         isAdminLogin = false;
738     }
739 }
740
741 public static void editQuiz() {
742     char wantToAddQuestion;
743     System.out.println("-----<<< EDIT QUIZ >>>-----");
744     System.out.println("-----");
745     System.out.println("SUBJECTS |");
746     subjects();
747     System.out.println("-----");
748
749     System.out.print("Enter Subject Name to Edit Quiz: ");
750     String quizSelection = input.nextline();
751
752     // Prompt admin to select a subject to edit
753     System.out.println("-----");
754     System.out.println("1: Easy |");
755     System.out.println("2: Medium |");
756     System.out.println("3: Hard |");
757     System.out.println("-----");
758
759     int difficultyOption;
760     // Prompt admin to select difficulty level for editing quiz
761     while (true) {
762         try {
763             System.out.println("Select a difficulty: ");
764             difficultyOption = input.nextInt();
765             if (difficultyOption < 1 || difficultyOption > 3) {
766                 System.out.println("Kindly Select a Valid Option");
767             }
768             else {
769                 break;
770             }
771         } catch (InputMismatchException e) {
772             System.out.println("Only Integers Are Accepted.....Try Again");
773         }
774     }
775
776     String difficulty = "";
777     switch (difficultyOption) {
778         case 1 -> difficulty = "Easy";
779         case 2 -> difficulty = "Medium";
780         case 3 -> difficulty = "Hard";
781     }
782
783     String subjectName = quizSelection + difficulty;
784
785     try {
786         //creating file writer object to edit the question file
787         FileWriter questionWriter = new FileWriter(subjectName + "Quiz.txt", append:true);
788         FileWriter keyWriter = new FileWriter(subjectName + "Key.txt", append:true);
789     }
790
791     //prompting admin to enter the question to add
792     System.out.println("Enter the Questions that you want to add along with the options: ");
793     System.out.println("In format: Q5 STATEMENT |A. |B. |C. |D. |");
794     input.nextline(); // consume leftover newline
795     String question = input.nextline();
796     //storing questions in the file
797     questionWriter.write(question + "\n");
798     //prompting admin to enter the key for the asked question
799     System.out.print("Now Enter the Key: ");
800     char key = input.next().toUpperCase().charAt(index:0);
801     //clearing buffer
802     input.nextline();
803     //storing answer in the key file
804     keyWriter.write(key + "\n");
805
806     //asking admin if want to add more question , otherwise exit from loop
807     System.out.print("Do You want to Add More Questions?(Y/N): ");
808     wantToAddQuestion = input.next().toUpperCase().charAt(index:0);
809     while (wantToAddQuestion == 'Y');
810     //closing resources

```

```

810 //closing resources
811 System.out.println("***** Questions Added Successfully! *****");
812 newLineWriter.close();
813 keyWriter.close();
814 }
815 catch (FileNotFoundException e){
816     System.out.println("File Does Not exist");
817 }
818 catch (IOException e){
819     System.out.println("Error while writing to the file");
820 }
821 }
822 char adminName;
823 System.out.print("Do You Want to Go to back to MENU then press Y otherwise " +
824     "press any other key to exit: ");
825 adminName = input.next().toUpperCase().charAt(index*10);
826 if (adminName != 'Y'){
827     System.out.println("Logging Out.....");
828     isAdminLogin = false;
829 }
830 }
831 public static String difficultyLevel() {
832     System.out.println("-----");
833     System.out.println("1: Easy");
834     System.out.println("2: Medium");
835     System.out.println("3: Hard");
836     System.out.println("-----");
837 }
838 int optionSelected = 0;
839 while (true) {
840     try {
841         System.out.print("Kindly Select a Difficulty Level[1-4]: ");
842         optionSelected = input.nextInt();
843         // getting out of loop if input is valid
844         if (optionSelected >= 1 && optionSelected <= 4) {
845             break;
846         } else {
847             System.out.print("Kindly Select Option 1, 2, 3, or 4: ");
848         }
849     } catch (InputMismatchException e) {
850         System.out.println("Only Integers are Accepted....Try Again");
851         input.nextLine(); // clear invalid input
852     }
853 }
854 String difficultySelected = "";
855 switch (optionSelected) {
856     case 1 -> difficultySelected = "Easy";
857     case 2 -> difficultySelected = "Medium";
858     case 3 -> difficultySelected = "Hard";
859     case 4 -> difficultySelected = "Surprise";
860 }
861 return difficultySelected;
862 }
863 }
864 public static void userLoginMenu(){
865     System.out.println("-----");
866     System.out.println("<< Welcome to User Login Section >>");
867     System.out.println("-----");
868     System.out.println("1: Register");
869     System.out.println("2: Login");
870     System.out.println("-----");
871     //variable that will store user's choice
872     int userOption;
873     //loop that will run till valid input is not given
874     while (true){
875         try {
876             System.out.print("Kindly Select an Option: ");
877             userOption = input.nextInt();
878             System.out.println("-----");
879             //clearing buffer
880             input.nextLine();
881             //getting out of loop if got the desired input
882             break;
883         } catch (InputMismatchException e){
884             System.out.println("Only Integers are Accepted..... Try Again");
885             input.nextLine();
886         }
887     }
888     //checking which option user selected from the USER-MENU
889     switch (userOption) {
890         case 1 -> {
891             registerUser();
892             if (age < 15){
893                 input.nextLine();
894                 System.out.println("You Can't Take This Quiz");
895                 System.out.println("Good Bye!");
896                 System.out.print("Press Enter to Exit: ");
897                 input.nextLine();
898                 return;
899             }
900         } else {
901             else {
902                 System.out.println("***** Kindly Enter Details to Now Login *****");
903                 //After registration prompting user to login
904                 loginUser();
905             }
906         }
907     }
908     //If user login successfully then show userQuizMenu
909     if (isAdminLogin) {
910         showUserQuizMenu();
911     }
912 }
913 public static void adminLoginMenu(){
914     System.out.println("-----");
915     System.out.println("<< Welcome to Admin Login Section >>");
916     System.out.println("-----");
917     loginAdmin();
918     while (isAdminLogin) {
919         showAdminQuizMenu();
920     }
921 }

```

Run | Debug

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⇒ Main Menu:

```

/=====\
|         <<  QUIZ  APP  >>         |
|         =====                 |
|         |  WELCOME  |             |
|         =====                 |
|                                     |
|-----|
|         +-----+                 |
|         |  1. USER  |             |
|         |  2. ADMIN |             |
|         +-----+                 |
|                                     |
|-----|
\=====/

```

>> Select an Option[1-2]:

⇒ **Login Section:**

```
>> Select an Option[1-2]: 2
=====
|      << Welcome to Admin Login Section >>      |
=====
>> UserName: Zabi
>> Password: Zabi@0000
***** Admin Login Successfully *****
```

⇒ Admin Panel:

```
=====
|      << Welcome to Admin Panel >>      |
|=====|
| 1: Add Quiz                             |
|-----|
| 2: View All Results                     |
|-----|
| 3: Edit Quiz                           |
|-----|
| 4: Delete Quiz                         |
|-----|
| 5: Exit                               |
|=====|
>> Kindly Select an option: |
```

⇒ Add Quiz :

```
>> Kindly Select an option: 1

>> Enter the Subject Name to Add: Civics
=====
| 1: EASY                               |
| 2: MEDIUM                           |
| 3: HARD                              |
|=====|
>> Select a Difficulty Level for Civics2
Civics- Added to the Subjects Successfully!
=> Quiz File CivicsMEDIUMQuiz.txt Created Successfully
=> Key File CivicsMEDIUMKey.txt Created Successfully
> Do You Want to Go to back to MENU then press Y otherwise press any other key to exit: |
```

⇒ View All Result:

```
=====
>> Kindly Select an option: 2
Zabi scored 7/20 in Surprise
Zabi scored 3/20 in EnglishEASY
Abdul moiz Israr scored 0/20 in Surprise
Abdul moiz Israr scored 7/20 in Surprise
Taha scored 6/20 in MathematicsHARD
>> Do You Want to Go to back to MENU then press Y otherwise press any other key to exit: |
```

⇒ Delete Quiz:

```
>> Kindly Select an option: 4
>> Enter the Subject Name to Delete: Civics
=====
| 1: EASY |
| 2: MEDIUM |
| 3: HARD |
=====
> Select a Difficulty Level for: Civics
2
All the files of Civics deleted Successfully!
Civics => Subject removed from the Subject File Successfully
```

➤ User Portal:

⇒ User Login Section:

```
>> Select an Option[1-2]: 1
=====
| << Welcome to User Login Section >> |
=====
| 1: Register |
| ----- |
| 2: Login |
=====
>> Kindly Select an Option: █
```

⇒ User Register:

```
>> Kindly Select an Option: 1

>> Enter Your Name: Haris
>> Enter Your Age: 19
>> Enter a strong Password: Haris@123
**** <Registered Successfully> ****
**** Kindly Enter Details to Now Login ****
>> UserName: █
```

⇒ User Login:

```
>> Kindly Select an Option: 2

>> UserName: Zabi
>> Age: 19
>> Password: Zabi@123
***** User Login Successful *****
=====
| << Welcome to User Quiz Section >> |
=====
| 1: Take Quiz |
| ----- |
| 2: View Results |
| ----- |
| 3: Exit |
=====
>> Kindly Select an Option: █
```

⇒ User Quiz Menu:

```
***** User Login Successful *****  
=====   
|    << Welcome to User Quiz Section >>    |   
=====   
| 1: Take Quiz                               |   
| -----|   
| 2: View Results                           |   
| -----|   
| 3: Exit                                   |   
=====   
>> Kindly Select an Option: |
```

⇒ Take Quiz:

⇒ Difficulty Level:

```
>> Kindly Select an Option: 1  
=====   
| 1: Easy                                   |   
| 2: Medium                               |   
| 3: Hard                                   |   
=====   
>> Kindly Select a Difficulty Level[1-4]: 1
```

⇒ Quiz Menu:

```
>> Kindly Select a Difficulty Level[1-4]: 1  
===== QUIZ MENU =====   
| 1: English                               |   
| 2: Urdu                                   |   
| 3: Mathematics                           |   
| 4: Physics                               |   
| 5: Computer                             |   
=====   
Enter the Subject Name to Attempt Quiz: |
```

⇒ START QUIZ:

```
Enter the Subject Name to Attempt Quiz: English
===== NOTE! =====
| You Have 2 minutes and 30 seconds to complete the Quiz |
=====
Q1. Choose the correct word: The sun rises in the _____.
A. west
B. north
C. south
D. east
Enter Your Answer: a
Q2. Select the correct word: We ____ happy today.
A. is
B. am
C. are
D. be
Enter Your Answer: b
Q3. Fill in the blank: He is ____ boy.
A. a
B. an
C. the
D. in
Enter Your Answer: a
Q4. Choose the opposite of 'Hot':
A. Warm
B. Cold
C. Sunny
D. Cool
Enter Your Answer: c
Q5. Select the correct spelling:
A. Apple
B. Applle
C. Aple
D. Appel
Enter Your Answer: a
Q6. Choose the correct sentence:
A. I am a student.
B. I student am.
C. Am I student.
D. Student am I.
Enter Your Answer: |
```

So on.....At end

```
Q20. Which is a bird?
A. Lion
B. Dog
C. Sparrow
D. Elephant
Enter Your Answer: a
You Scored 7 out of 20
Do You Want to Go to back to MENU then press Y otherwise press any other key to exit
```

⇒ View User Own Result:

```
>> Kindly Select an Option: 2
Zabi Scored 4 out of 20 in ComputerQuiz
Zabi Scored 6 out of 20 in UrduQuiz
Zabi Scored 7 out of 20 in UrduQuiz
Zabi Scored 6 out of 20 in MathematicsQuiz
Zabi Scored 7 out of 20 in SurpriseQuiz
Zabi Scored 3 out of 20 in EnglishEASYQuiz
Zabi Scored 7 out of 20 in EnglishEASYQuiz
>> Do You Want to Go to back to MENU then press Y otherwise press any other key to exit: |
```

⇒ Exit:

```
=====
|   << Welcome to User Quiz Section >>   |
=====
|  1: Take Quiz                            |
|  -----                                |
|  2: View Results                         |
|  -----                                |
|  3: Exit                                |
=====
>> Kindly Select an Option: 3
> Logging Out.....
PS D:\Code\JAVA\Final semster Project> |
```

THE END